Teacher answer key: Primary 6, Semester 1 (DRAFT)

DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, with the outcome numbers of the Primary 6 Standard content in the Mathematics Syllabus, Primary (2021 edition, pages 43 and 44). The outcome wording in this booklet is a summary in our own words, not the syllabus text; where the summary gave no outcome number the key gives none. The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

Pages in this booklet (Terms 1 and 2)

Strand	Pages
Number and Algebra - Fractions	S1-01, S1-02, S1-03
Number and Algebra - Ratio	S1-04, S1-05, S1-06, S1-07, S1-08
Number and Algebra - Ratio and Percentage	S1-09
Number and Algebra - Percentage	S1-10, S1-11, S1-12
Number and Algebra - Algebra	S1-13, S1-14, S1-15, S1-16
Measurement and Geometry - Area and Volume	S1-17, S1-18, S1-19, S1-20

The page code in the footer, for example S1-07 (P6-07), gives the page in this booklet and, in brackets, its number in the full set of 40.

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S1-01 (P6-01) Fraction Shares 1 star

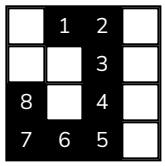
Topic: Dividing a fraction by a whole number

Syllabus outcomes: Fractions 1.1. Notes on the tag: 1.1 (dividing a proper fraction by a whole number)
Puzzle type: Code-breaker (fractions). How the pupil checks: Code word (6 letters).
Answers: 1) 2/25; 2) 3/13; 3) 1/27; 4) 2/15; 5) 5/27; 6) 3/56
Check result: Code word: CREAMY
Teacher note: Six items: four divisions of a proper fraction by a whole number (the top divides exactly, the top does not divide, the answer needs simplifying, a unit fraction) and two stories. The decoy letters stand for the answers a pupil gets by multiplying instead of dividing, by dividing the bottom number, or by leaving the answer unsimplified. A box left blank gives no letter and the check cannot see it.

S1-02 (P6-02) Dividing by Fractions 1 star

Topic: Dividing a whole number or a fraction by a fraction

Syllabus outcomes: Fractions 1.2. Notes on the tag: 1.2 (dividing a whole number or a proper fraction by a proper fraction)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box equal to 6.
Answers: Route (row, column from top-left): (1,2) 2 ÷ ⅓ -> (1,3) ⁶⁄₁₁ ÷ ¹⁄₁₁ -> (2,3) ½ ÷ ¹⁄₁₂ -> (3,3) 1 ÷ ⅙ -> (4,3) ⅔ ÷ ⅑ -> (4,2) 4 ÷ ⅔ -> (4,1) 5 ÷ ⅚ -> (3,1) ¾ ÷ ⅛
Check result: Route reaches FINISH through 8 boxes; every box on it equals 6.



Teacher note: A 4 by 4 grid of sixteen divisions: a whole number or a proper fraction divided by a proper fraction. The route follows the boxes that equal 6. The wrong boxes include answers close to 6 and boxes where the two numbers multiply to 6, which tempts a pupil who multiplies instead of dividing. Two extra boxes that equal 6 are dead ends. Counting the boxes shows a line that takes a shortcut.

S1-03 (P6-03) Fraction Stories 2 stars

Topic: Fraction stories: dividing by a whole number or by a fraction

Syllabus outcomes: Fractions 1.1, Fractions 1.2. Notes on the tag: 1.1, 1.2 (dividing a proper fraction by a whole number; dividing by a proper fraction)
Puzzle type: Choice table (stories). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 2; 2) ²⁄₉; 3) 9; 4) ¹⁄₁₀; 5) 3; 6) 5
Check result: Word: MONKEY
Teacher note: Six stories: how many pieces of a given fraction of a metre fit into a whole number of metres, what fraction of a whole each child gets when a fraction is shared, and how many glasses of a fraction of a litre a jug fills. Each row has three answers. The wrong answers are what a pupil gets by multiplying instead of dividing, or by dividing the other way round. The letters of the right answers, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

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S1-04 (P6-04) Equal Ratios 1 star

Topic: Ratios: equivalent ratios and simplest form

Syllabus outcomes: Ratio 1.1, Ratio 1.2, Ratio 1.4. Notes on the tag: 1.1, 1.2, 1.4 (ratio a : b and a : b : c; equivalent ratios; simplest form)
Puzzle type: Matching (equal ratios). How the pupil checks: Letters of the matching cards, read in the order of the items, spell a word (6 letters).
Answers: 1) card S; 2) card H; 3) card R; 4) card I; 5) card M; 6) card P
Check result: Word: SHRIMP
Teacher note: Six ratios (four with two parts, two with three parts) are matched to eight cards. A card is the item in simplest form or multiplied by a different number, so the pupil must simplify or scale. Two cards are decoys: a ratio with the parts written the other way round, or with a part changed. The letters of the matching cards, read in the order of the items, spell the word. Every way of giving each item a different card, and every single changed letter, was tested against a very large word list: none spells another word. A box left blank gives no letter and the check cannot see it.

S1-05 (P6-05) Missing Terms 1 star

Topic: Equivalent ratios: finding the missing term

Syllabus outcomes: Ratio 1.2, Ratio 1.6. Notes on the tag: 1.2, 1.6 (equivalent ratios; finding the missing term in equivalent ratios)
Puzzle type: Code-breaker (equivalent ratios). How the pupil checks: Code word (6 letters).
Answers: 1) 48; 2) 15; 3) 9; 4) 54; 5) 27; 6) 44
Check result: Code word: TEACUP
Teacher note: Six items: three two-part equations with the missing number in a different place, one three-part equation and two sharing stories. The decoy numbers are the answers a pupil gets by adding the same number to both terms instead of multiplying. A box left blank gives no letter and the check cannot see it.

S1-06 (P6-06) Share in a Ratio 1 star

Topic: Ratio: sharing a quantity in a given ratio

Syllabus outcomes: Ratio 1.3, Ratio 1.5. Notes on the tag: 1.3, 1.5 (dividing a quantity in a given ratio; ratio of two or three quantities)
Puzzle type: Code-breaker (sharing in a ratio). How the pupil checks: Code word (6 letters).
Answers: 1) 12; 2) 16; 3) 20; 4) 30; 5) 147; 6) 70
Check result: Code word: BANDIT
Teacher note: Six items: the larger share, the smaller share, how much more one share is than the other, the largest share of a three-part ratio, the shortest piece of a rope, and one part of a total given as a ratio of two groups. The decoy numbers are what a pupil gets by dividing the total by the number of parts only, by finding the wrong share, or by taking the other part. A box left blank gives no letter and the check cannot see it.

S1-07 (P6-07) Three Quantities 2 stars

Topic: Ratio of three quantities in simplest form

Syllabus outcomes: Ratio 1.1, Ratio 1.4, Ratio 1.5. Notes on the tag: 1.1, 1.4, 1.5 (ratio a : b : c; simplest form; ratio of two or three quantities)
Puzzle type: Code-breaker (ratio of three quantities). How the pupil checks: Code word (6 letters).
Answers: 1) 2 : 7 : 4; 2) 7 : 5 : 1; 3) 5 : 6 : 2; 4) 1 : 2 : 2; 5) 6 : 4 : 3; 6) 4 : 1 : 3
Check result: Code word: DEVOUR
Teacher note: Six items: three counts, two quantities with units that must first be changed to the same unit, and two linked ratios (the middle quantity appears in both). The decoy ratios are what a pupil gets by leaving the ratio unsimplified, by writing the quantities in the wrong order, by not changing the units, or by not matching the middle term of two linked ratios. A box left blank gives no letter and the check cannot see it.

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S1-08 (P6-08) Fractions and Ratios 2 stars

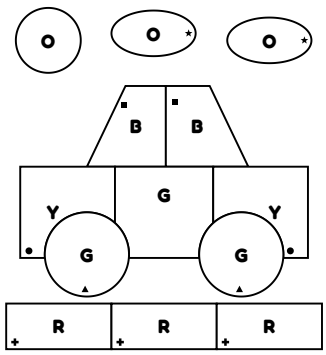
Topic: Ratio and fraction: how they are linked

Syllabus outcomes: Ratio 1.7, Ratio 1.5. Notes on the tag: 1.7 (relationship between fraction and ratio); 1.5
Puzzle type: Choice table (ratio and fraction). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 4 : 1; 2) 2 : 7; 3) ⅙; 4) ⁷⁄₁₂; 5) ³⁄₈; 6) 3 : 4
Check result: Word: FATHOM
Teacher note: Six questions that link a fraction and a ratio: a fraction of a quantity to a ratio of the part to the rest, a ratio to the fraction one part makes of the whole, a fraction taken as a ratio of two quantities, and one part as a fraction of the other part. Each row has three answers. The wrong answers are what a pupil gets by using the whole instead of the rest, by writing the parts the wrong way round, or by writing a ratio as a fraction of the other part. The letters of the right answers, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

S1-09 (P6-09) Colour the Car 2 stars

Topic: Ratio and percentage

Syllabus outcomes: Ratio 1.2, Ratio 1.3, Ratio 1.6, Percentage 1.1, Percentage 1.2. Notes on the tag: Ratio 1.2, 1.3, 1.6 and Percentage 1.1, 1.2 (equivalent ratios, sharing in a ratio, the missing term, finding the whole, percentage change)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and parts with the same small mark match.
Answers: Colour key: 12 = yellow, 20 = red, 36 = blue, 45 = green, 60 = orange
Check result: Tally: yellow 2, red 3, blue 2, green 3, orange 3; parts with the same small mark match.



Teacher note: Every part asks a ratio or percentage question with a whole-number answer: a missing term in equal ratios, a share of a quantity shared in a ratio, a percentage of a number, the whole when a percentage of it is given, or the number before a percentage increase. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); parts with the same small mark match.

S1-10 (P6-10) Find the Whole 1 star

Topic: Percentage: finding the whole from a part

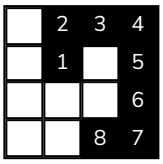
Syllabus outcomes: Percentage 1.1. Notes on the tag: 1.1 (finding the whole given a part and the percentage)
Puzzle type: Code-breaker (percentage). How the pupil checks: Code word (6 letters).
Answers: 1) 640; 2) 160; 3) 190; 4) 660; 5) 360; 6) 480
Check result: Code word: FRUITY
Teacher note: Six questions in which a part and its percentage are given and the whole is wanted: a bare number, pupils in a hall, a sale price, savings, a length and a tank. The decoy numbers are what a pupil gets by finding the percentage of the part instead, by adding the percentage on to the part, or by subtracting the part from the whole. A box left blank gives no letter and the check cannot see it.

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S1-11 (P6-11) Percent Up and Down 1 star

Topic: Percentage increase and decrease

Syllabus outcomes: Percentage 1.2. Notes on the tag: 1.2 (percentage increase or decrease)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box equal to a 25% increase.
Answers: Route (row, column from top-left): (2,2) Up from 16 to 20 -> (1,2) Up from 88 to 110 -> (1,3) Up from 220 to 275 -> (1,4) Up from 116 to 145 -> (2,4) Up from 8 to 10 -> (3,4) Up from 120 to 150 -> (4,4) Up from 224 to 280 -> (4,3) Up from 236 to 295
Check result: Route reaches FINISH through 8 boxes; every box on it equals a 25% increase.



Teacher note: A 4 by 4 grid of sixteen changes, each written as Up or Down from one number to another. The route follows the boxes that are a 25% increase. The wrong boxes include 25% decreases (a pupil who ignores the word Down) and increases from a smaller number to the old one (a rise of one third, not 25%), and other percentages. Two extra 25% increases are dead ends. Counting the boxes shows a line that takes a shortcut.

S1-12 (P6-12) Percent Stories 2 stars

Topic: Percentage: finding the whole and the percentage change

Syllabus outcomes: Percentage 1.1, Percentage 1.2. Notes on the tag: 1.1, 1.2 (finding the whole given a part and the percentage; percentage increase or decrease)
Puzzle type: Code-breaker (percentage stories). How the pupil checks: Code word (6 letters).
Answers: 1) 5; 2) 400; 3) 30; 4) 420; 5) 320; 6) 340
Check result: Code word: SQUARE
Teacher note: Six stories: the usual price after a discount, the old price after a rise, the amount at first after some were sold, a percentage increase, a percentage decrease, and the smaller quantity when the other is a percentage more. Three need the whole from a part that is a percentage above or below it, so the pupil must divide by the right percentage. The decoy numbers are what a pupil gets by taking the percentage of the given amount, by finding only the difference, or by adding or subtracting the percentage as a number. A box left blank gives no letter and the check cannot see it.

S1-13 (P6-13) Letters for Numbers 1 star

Topic: Algebra: writing expressions with a letter

Syllabus outcomes: Algebra 1.1, Algebra 1.2. Notes on the tag: 1.1, 1.2 (a letter for an unknown number; notation of simple expressions such as a + 3, 3a and a ÷ 3)
Puzzle type: Choice table (expressions). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) k ÷ 3; 2) n + 4; 3) 6 y; 4) t – 7; 5) p – 9; 6) 8 w
Check result: Word: STUDIO
Teacher note: Six short stories in which a letter stands for an unknown number: a price that is more, a number left after some are taken, the cost of several equal items, a share among friends, an age that is younger, and a number of items in several boxes. The pupil chooses the right expression of three. The wrong expressions use the wrong operation or write the number and the letter in the wrong order (for example K ÷ n for n ÷ K). The notation is the usual one: 3n for n times 3 and n ÷ 3. The letters of the right choices, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

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S1-14 (P6-14) Simplify 1 star

Topic: Algebra: simplifying expressions with one letter

Syllabus outcomes: Algebra 1.3. Notes on the tag: 1.3 (simplifying simple linear expressions, no brackets)
Puzzle type: Matching (simplest form). How the pupil checks: Letters of the matching cards, read in the order of the items, spell a word (6 letters).
Answers: 1) card W; 2) card A; 3) card L; 4) card R; 5) card U; 6) card S
Check result: Word: WALRUS
Teacher note: Six expressions with one letter and no brackets: adding like terms, subtracting like terms, and expressions where the letter terms and the numbers are mixed up, one with a lone letter (n means 1n). Each is matched to its simplest form among eight cards. Two cards are decoys: a wrong simplification, such as the two numbers swapped or all the terms added together. Every way of giving each item a different card, and every single changed letter, was tested against a very large word list: none spells another word. A box left blank gives no letter and the check cannot see it.

S1-15 (P6-15) Substitute 2 stars

Topic: Algebra: finding the value of an expression

Syllabus outcomes: Algebra 1.4, Algebra 1.2. Notes on the tag: 1.4 (evaluating a simple expression by substitution)
Puzzle type: Domino loop (substituting). How the pupil checks: Loop: the right half of each domino equals the left half of the next, all the way round to START; the letters in loop order spell a word (6 letters).
Answers: Loop from START: B (5 k + 18 when k = 7 | w ÷ 4 + 35 when w = 20) -> O (a ÷ 4 + 31 when a = 36 | k ÷ 2 + 8 when k = 18) -> T (k ÷ 4 + 13 when k = 16 | 4 t + 12 when t = 9) -> A (3 p + 27 when p = 7 | 2 p + 4 when p = 7) -> N (5 p − 12 when p = 6 | t ÷ 2 + 34 when t = 10) -> Y (t ÷ 3 + 35 when t = 12 | 5 k + 23 when k = 6)
Check result: Word: BOTANY
Teacher note: Six dominoes. Each half shows an expression and the number its letter stands for. The right half of one domino has the same value as the left half of exactly one other domino, so the dominoes make one loop that comes back to START. The pupil finds the value of each half, follows the loop from START and writes the letters in loop order; they spell the word. All twelve values are in two groups of six equal pairs, and every wrong order was tested against a very large word list: none spells a word.

S1-16 (P6-16) Find the Letter 2 stars

Topic: Algebra: solving simple equations

Syllabus outcomes: Algebra 1.5, Algebra 1.3. Notes on the tag: 1.5 (simple linear equations with whole-number coefficients)
Puzzle type: Code-breaker (equations). How the pupil checks: Code word (6 letters).
Answers: 1) w = 6; 2) p = 26; 3) k = 2; 4) n = 29; 5) y = 3; 6) t = 23
Check result: Code word: CRAFTY
Teacher note: Six equations with one letter and whole-number coefficients: one step with addition, one with subtraction, one with a coefficient only, one that first needs like terms collected, and two stories that the pupil turns into an equation. The decoy numbers are what a pupil gets by adding instead of subtracting the number, by forgetting to divide, or by dividing before subtracting. A pupil can test any answer by putting it back in the equation. A box left blank gives no letter and the check cannot see it.

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S1-17 (P6-17) Round and Round 1 star

Topic: Circles: circumference

Syllabus outcomes: Area and Volume 1.1. Notes on the tag: 1.1 (area and circumference of a circle)
Puzzle type: Code-breaker (circumference). How the pupil checks: Code word (6 letters).
Answers: 1) 176 cm; 2) 88 cm; 3) 154 cm; 4) 22 m; 5) 66 cm; 6) 44 cm
Check result: Code word: COUSIN
Teacher note: Six circles: three drawn, with the diameter or the radius marked, and three stories. Every diameter is a multiple of 7, so with π as 22/7 every answer is a whole number. The pupil has to use the diameter or double the radius. The decoy numbers are what a pupil gets by finding the area, by using the radius instead of the diameter, or by halving. A box left blank gives no letter and the check cannot see it.

S1-18 (P6-18) Circle Areas 2 stars

Topic: Circles: area

Syllabus outcomes: Area and Volume 1.1. Notes on the tag: 1.1 (area and circumference of a circle)
Puzzle type: Code-breaker (area). How the pupil checks: Code word (6 letters).
Answers: 1) 50.24; 2) 153.86; 3) 200.96; 4) 314; 5) 12.56; 6) 28.26
Check result: Code word: FROZEN
Teacher note: Six circles: two drawn (one with the radius, one with the diameter), three stories and one two-step item that gives the circumference, so the pupil first finds the radius. Every radius is a whole number from 2 to 10, so every answer has at most two decimal places. The decoy numbers are what a pupil gets by using $2 \times \pi \times r$ (the circumference) or $\pi \times r$ instead of $\pi \times r \times r$, or by squaring the diameter. A box left blank gives no letter and the check cannot see it.

S1-19 (P6-19) Half and Quarter 2 stars

Topic: Semicircles and quarter circles: area and perimeter

Syllabus outcomes: Area and Volume 1.2. Notes on the tag: 1.2 (area and perimeter of a semicircle and a quarter circle)
Puzzle type: Matching (areas and perimeters). How the pupil checks: Letters of the matching cards, read in the order of the items, spell a word (6 letters).
Answers: 1) card H; 2) card U; 3) card N; 4) card G; 5) card R; 6) card Y
Check result: Word: HUNGRY
Teacher note: Three semicircles and three quarter circles, three asking for the area and three for the perimeter, with sizes that are multiples of 7 so that every answer is exact. Two cards are decoys: a card with the number of a perimeter but the unit cm^2, and the length of the curved edge alone. The perimeter of a semicircle or quarter circle includes its straight edges. The letters of the matching cards, read in the order of the items, spell the word. Every way of giving each item a different card, and every single changed letter, was tested against a very large word list: none spells another word. A box left blank gives no letter and the check cannot see it.

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S1-20 (P6-20) Curves and Corners 2 stars

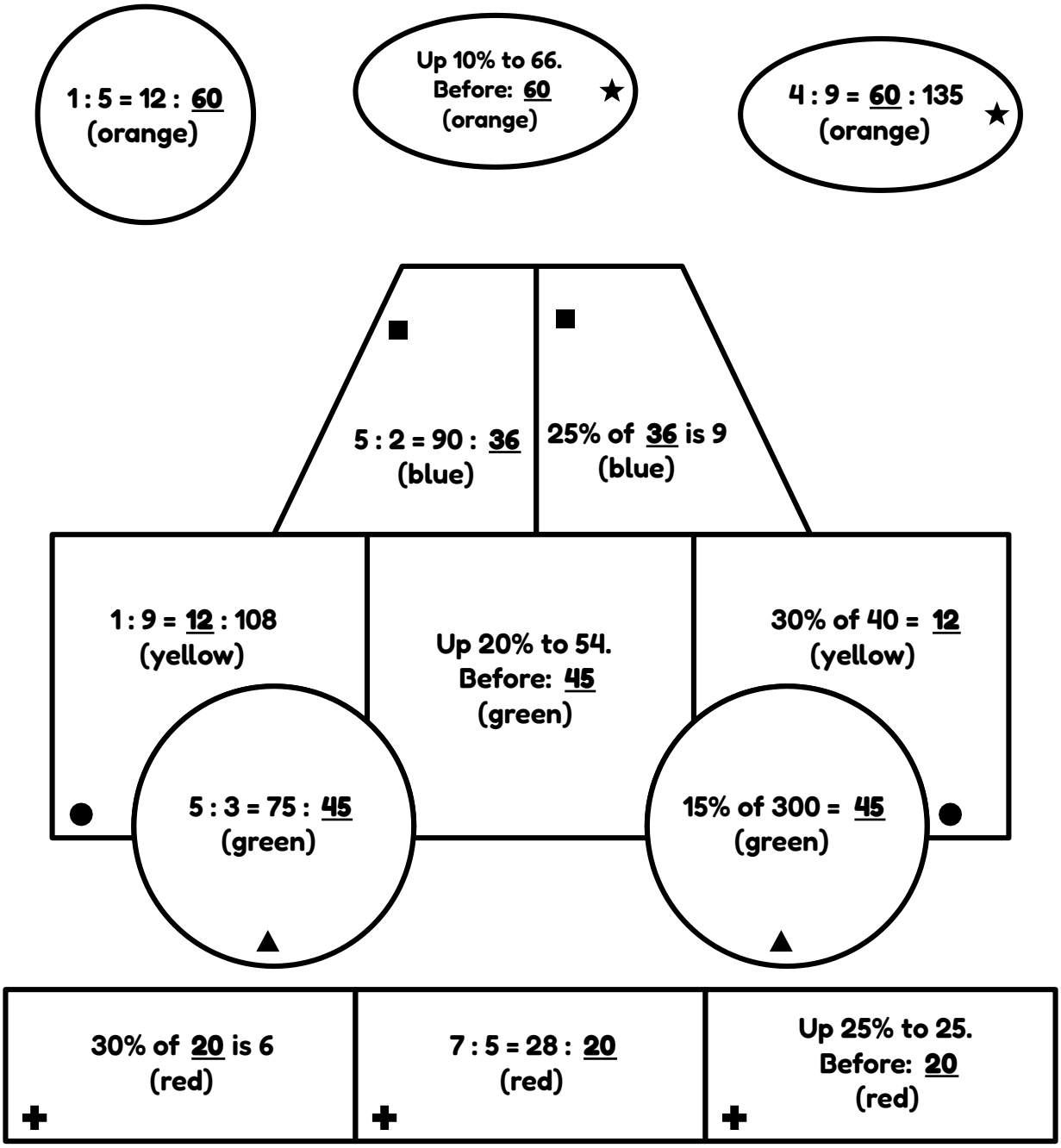
Topic: Composite figures with semicircles and quarter circles

Syllabus outcomes: Area and Volume 1.3, Area and Volume 1.2. Notes on the tag: 1.3 (composite figures made of squares, rectangles, semicircles and quarter circles); 1.1
Puzzle type: Ordering (areas of composite figures). How the pupil checks: Letters of the cards in the order asked spell a word (6 letters).
Answers: smallest area first: P = 42 cm² (square less quarter circle 14), O = 68.25 cm² (square and semicircle 7), L = 94.5 cm² (square less quarter circle 21), I = 371 cm² (rectangle and semicircle 21x14), C = 385 cm² (rectangle and semicircle 22x14), E = 614.25 cm² (square and semicircle 21)
Check result: Word: POLICE
Teacher note: Six figures: a rectangle with a semicircle on one end, a square with a semicircle on one side, and a square with a quarter circle cut away (the hatched part). All sizes are chosen so that the areas are exact. The areas of the six figures are all different and differ by at least 6. The letters, read from the smallest area to the greatest, spell the word; every wrong order was tested against a very large word list and none spells a word. A box left blank gives no letter and the check cannot see it.

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Answers: S1-09 (P6-09) Colour the Car

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 12 = yellow, 20 = red, 36 = blue, 45 = green, 60 = orange.



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